

Psychographic Segmentation: Gender-Based Analysis of Personality Traits

Purpose & Big Five

Survey Results

Chi-Square Test

Mann-Whitney U Test

Consensus & Profiling

Big Five Personality Traits

Objective: To conduct a comparative analysis of male and female personality profiles among the 18 to 26 age cohort in India, utilizing the **Big Five Personality Trait model**.

Goal: The study aims to identify similarities, key differences, and which gender exhibits the highest average scores across the five core personality dimensions.

Tools Used: The analysis employed **Excel**, **SPSS**, and **Tableau** for data processing, statistical analysis, and visualization.

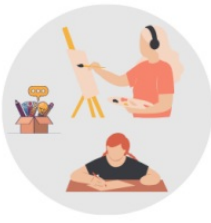
Data Collection: Participants completed a [60-item online questionnaire](#) using a 5-point Likert scale, assessing 12 traits per Big Five dimension. To control response bias, negatively-keyed items were reverse-coded.

Consensus Analysis: The Coefficient of Variation (C.V.) was computed based on weighted response frequencies to create a standardized measure of consensus (agreement) within each gender.

Chi-Square Test of Independence: To find if there is a statistical association between Gender and the specific trait statements.

Mann-Whitney U Test: To determine if the overall distribution of Likert responses significantly differs between genders.

The Big Five Personality Trait Model

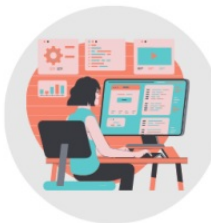


Openness to Experience

reflects an individual's willingness to try new things and embrace unconventional ideas.

People scoring high are intellectually curious, sensitive to beauty, and willing to challenge their existing notions.

Conversely, those scoring low are often pragmatic, closed-minded, and wary of uncertainty, preferring familiar routines, traditions, and environments.



Conscientiousness reflects an individual's tendency toward self-discipline, organization, and goal-oriented behavior.

Individuals with **high conscientiousness** are punctual, ambitious, and prefer planned behavior and tidy environments.

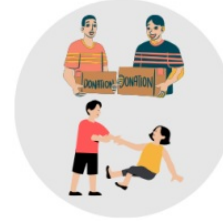
Low conscientiousness is associated with flexibility, spontaneity, and impulsive behavior, often resulting in less motivation and a more relaxed approach to setting and meeting goals.



Extraversion measures the intensity and preference for social engagement.

Extraverts seek external stimulation, thriving in the company of others as they are outgoing and sociable.

Introverts (low Extraversion) are quiet, reserved, and need less external stimulation, preferring time alone or in smaller groups. This trait is a continuum, and those in the middle are known as **Ambiverts**.



Agreeableness reflects an individual's orientation toward social harmony and cooperation.

People scoring high are friendly, kind, and trustworthy, typically holding an optimistic view of human nature and actively avoiding conflict.

Disagreeable individuals are often suspicious of others' intentions and behave primarily in their own self-interest, and being perceived as less charitable.



Neuroticism measures an individual's tendency to experience negative emotions like stress, anxiety, and anger.

Individuals with **high neuroticism** are emotionally reactive, over-think problems, and struggle to cope effectively with stress.

Conversely, those scoring **low in neuroticism** are emotionally stable, rarely upset, and able to remain calm and worry less when facing stressful situations.

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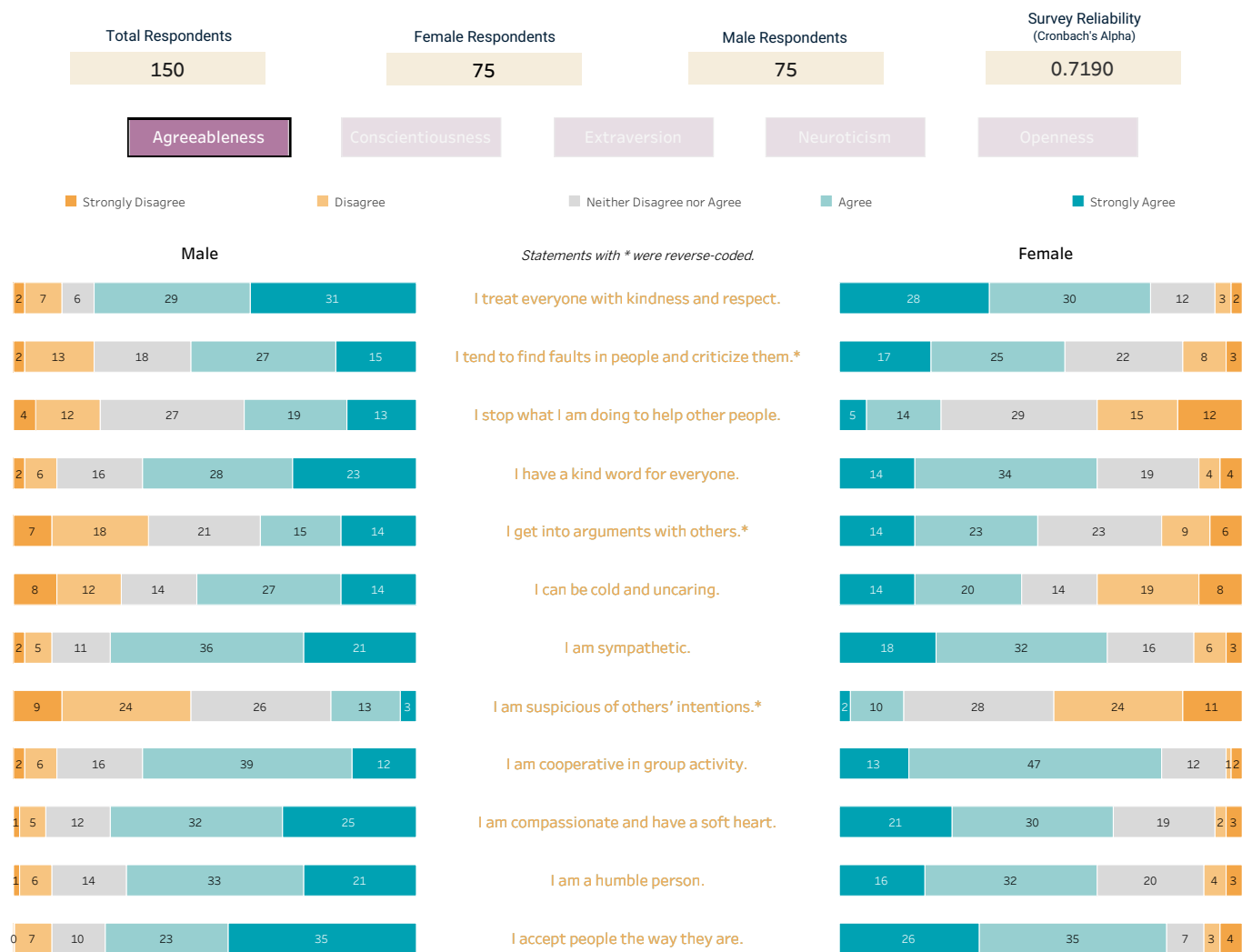
Chi-Square Test

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BIG FIVE PERSONALITY SURVEY ANALYSIS

The charts display the Likert scale results of every question for each of the **Big Five Personality** dimensions. The quality of the survey was checked using a reliability test. **Cronbach's Alpha** measures if the questions for each trait are accurate and well-designed. A higher Alpha value means the questions consistently measure the same dimension.



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CHI SQUARE TEST FOR GENDER ASSOCIATION

The purpose of conducting the test is to find out whether there is an association between gender and their response choices (Strongly Disagree - Strongly Agree) for each statements under the Big Five Dimensions. It reveals if the pattern of **responses** across all categories is different between genders.

H0: There is no significant association between Gender and the response to the chosen Dimension's statement.

H1: There is a significant association between Gender and the response to the chosen Dimension's statement.

Alpha (α) = 0.05

Agreeableness

Conscientiousness

Extraversion

Neuroticism

Openness

Statements	P Value	Conclusion	
I accept people the way they are.	0.041	Since the p-value is less than alpha, we reject the null hypothesis. Gender is a significant factor in the response to this statement.	Accept the Alternative Hypothesis
I am a humble person.	0.533	Since the p-value is greater than alpha, we fail to reject the null hypothesis. Thus, Gender is not a significant factor in the response to this statement.	Fail to Reject the Null Hypothesis
I am compassionate and have a soft heart.	0.37	Since the p-value is greater than alpha, we fail to reject the null hypothesis. Thus, Gender is not a significant factor in the response to this statement.	Fail to Reject the Null Hypothesis
I am cooperative in group activity.	0.295	Since the p-value is greater than alpha, we fail to reject the null hypothesis. Thus, Gender is not a significant factor in the response to this statement.	Fail to Reject the Null Hypothesis
I am suspicious of others' intentions.*	0.929	Since the p-value is greater than alpha, we fail to reject the null hypothesis. Thus, Gender is not a significant factor in the response to this statement.	Fail to Reject the Null Hypothesis
I am sympathetic.	0.794	Since the p-value is greater than alpha, we fail to reject the null hypothesis. Thus, Gender is not a significant factor in the response to this statement.	Fail to Reject the Null Hypothesis
I can be cold and uncaring.	0.623	Since the p-value is greater than alpha, we fail to reject the null hypothesis. Thus, Gender is not a significant factor in the response to this statement.	Fail to Reject the Null Hypothesis
I get into arguments with others.*	0.303	Since the p-value is greater than alpha, we fail to reject the null hypothesis. Thus, Gender is not a significant factor in the response to this statement.	Fail to Reject the Null Hypothesis
I have a kind word for everyone.	0.393	Since the p-value is greater than alpha, we fail to reject the null hypothesis. Thus, Gender is not a significant factor in the response to this statement.	Fail to Reject the Null Hypothesis
I stop what I am doing to help other people.	0.069	Since the p-value is greater than alpha, we fail to reject the null hypothesis. Thus, Gender is not a significant factor in the response to this statement.	Fail to Reject the Null Hypothesis
I tend to find faults in people and criticize them.*	0.737	Since the p-value is greater than alpha, we fail to reject the null hypothesis. Thus, Gender is not a significant factor in the response to this statement.	Fail to Reject the Null Hypothesis
I treat everyone with kindness and respect.	0.438	Since the p-value is greater than alpha, we fail to reject the null hypothesis. Thus, Gender is not a significant factor in the response to this statement.	Fail to Reject the Null Hypothesis

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MANN-WHITNEY U TEST

The purpose of conducting this test is to find out whether the distribution of Likert responses differs significantly between genders for each statements under the Big Five Dimensions. It reveals if one gender tends to score higher or lower overall on the statement than the other gender.

H0: The distributions of the responses for the two Gender groups are the same. (The median response score is the same for both genders.)

H1: The distributions of the responses for the two Gender groups are different. (The median response score is different for the two genders.)

Alpha (α) = 0.05

Agreeableness

Conscientiousness

Extraversion

Neuroticism

Openness

Statements	P Value	Conclusion	
I am full of energy.	0	Since the p-value is less than alpha, we reject the null hypothesis. There is a statistically significant difference in the median response between genders.	Accept the Alternative Hypothesis
I am good at handling social situations.	0.021	Since the p-value is less than alpha, we reject the null hypothesis. There is a statistically significant difference in the median response between genders.	Accept the Alternative Hypothesis
I am outgoing and sociable.	0	Since the p-value is less than alpha, we reject the null hypothesis. There is a statistically significant difference in the median response between genders.	Accept the Alternative Hypothesis
I am quiet.*	0.405	Since the p-value is greater than alpha, we fail to reject the null hypothesis. There is no statistically significant difference in the median response between genders.	Fail to Reject the Null Hypothesis
I am shy around people.*	0.048	Since the p-value is less than alpha, we reject the null hypothesis. There is a statistically significant difference in the median response between genders.	Accept the Alternative Hypothesis
I am talkative.	0.401	Since the p-value is greater than alpha, we fail to reject the null hypothesis. There is no statistically significant difference in the median response between genders.	Fail to Reject the Null Hypothesis
I feel comfortable around people.	0.002	Since the p-value is less than alpha, we reject the null hypothesis. There is a statistically significant difference in the median response between genders.	Accept the Alternative Hypothesis
I find it hard to influence people.*	0.862	Since the p-value is greater than alpha, we fail to reject the null hypothesis. There is no statistically significant difference in the median response between genders.	Fail to Reject the Null Hypothesis
I like being the center of attention.	0.009	Since the p-value is less than alpha, we reject the null hypothesis. There is a statistically significant difference in the median response between genders.	Accept the Alternative Hypothesis
I rarely feel excited or eager.*	0.495	Since the p-value is greater than alpha, we fail to reject the null hypothesis. There is no statistically significant difference in the median response between genders.	Fail to Reject the Null Hypothesis
I show a lot of enthusiasm.	0.005	Since the p-value is less than alpha, we reject the null hypothesis. There is a statistically significant difference in the median response between genders.	Accept the Alternative Hypothesis
I try to be dominant and take charge of situations or events.	0	Since the p-value is less than alpha, we reject the null hypothesis. There is a statistically significant difference in the median response between genders.	Accept the Alternative Hypothesis

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Gender Consensus Ranking and Average Trait Profile

Coefficient of Variation (C.V.) measures level of agreement (consensus) on a statement for each gender. **Low C.V. means more agreement among the respondents, suggesting the trait is highly characteristic of a particular gender.** The statements were ranked according to their C.V. The top traits across both groups have been highlighted, as well as the traits where respondents opinions diverge the most.

Agreeableness

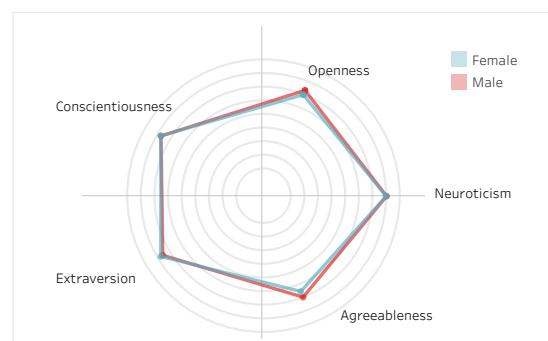
Conscientiousness

Extraversion

Neuroticism

Openness

Coefficient of Variation (C.V.)			Rank		
Statements	Female	Male	Statements	Female	Male
I always make good use of time.	83.30	99.68	I always make good use of time.	1	10
I avoid taking responsibility.*	83.64	96.21	I avoid taking responsibility.*	2	7
It's important to me that everyone should be on time.	85.36	85.30	It's important to me that everyone should be on time.	3	2
I often forget things.*	86.16	85.63	I often forget things.*	4	3
I am organized.	86.94	97.95	I am organized.	5	8
I find it difficult to get started on a task.*	89.67	94.00	I find it difficult to get started on a task.*	6	6
I get chores done right away.	93.87	128.28	I get chores done right away.	7	12
I am systematic.	96.79	78.97	I am systematic.	8	1
I make plans and stick to them.	99.04	88.55	I make plans and stick to them.	9	5
I often get lazy.*	104.47	86.58	I often get lazy.*	10	4
I am always prepared.	104.47	104.65	I am always prepared.	11	11
I am persistent.	117.22	99.60	I am persistent.	12	9



The chart shows that the overall shape of the personality profile is nearly identical for both genders.

The Male line appears to extend slightly further out on the Openness and Agreeableness axes. This suggests Males scored marginally higher in these two dimensions.

The Female line is slightly further out than the Male line on the Extraversion axis. This suggests that Females scored slightly higher on average in this dimension.

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